MGMTFT/MGMTFE 410: Operations & Technology Management
Practice Final Examination — Exam 2

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Expectations of ethical conduct

As a member of the UCLA Anderson community, I am guided in my academic pursuits by the principles of integrity, respect, and honesty. I strive to embody these principles, not because I must, but because I believe it is the right thing to do.

For this exam, it is a violation of the honor code to receive or provide assistance by any means. Do not communicate with other students about the exam during the exam period. Do not use generative AI tools (ChatGPT, Claude, etc.) during the exam.

By signing this document, I acknowledge my obligations under the UCLA Anderson Honor Code and certify that I will not give or receive assistance during this examination.

_____ _____

Student's Signature (or name) Date

_____

Print Name

_____ _____

Student ID Section

Instructions

1. Print your name and student UID on the cover page.
2. You may ONLY use your cheatsheet and a calculator. You MAY NOT use a phone calculator, laptop, or any other device. You may not use notes other than your cheatsheet.
3. **Various definitions and formulas that we covered in class are provided at the end of this exam.**
4. Write your answers in the spaces provided.
5. For the open questions, in order to receive partial credit, you must show all your work.
6. For the multiple-choice questions, you do not need to show your work, and no partial credit will be awarded.
7. State any assumptions that you make.

Good Luck!!!

Question 1: miscellaneous multiple-choice questions. [16 POINTS]

1. [2 points] A café has an average of 24 customers seated at any one time. On average, each customer stays 20 minutes. By Little's Law, what is the arrival rate?
   a. 0.4 customers/minute.
   b. 1.2 customers/minute.
   c. 480 customers/minute.
   d. Cannot be determined from the information given.

   Answer: _____

2. [2 points] In the EOQ model, the fixed ordering cost K doubles (all other parameters unchanged). What happens to the optimal order quantity Q*?
   a. It doubles.
   b. It increases by approximately 41% (multiplied by √2).
   c. It stays the same.
   d. It decreases by approximately 29% (multiplied by 1/√2).

   Answer: _____

3. [2 points] A three-step serial process has step capacities of 5, 8, and 3 units/hour. Current demand is 4 units/hour. What is the flow rate?
   a. 8 units/hour.
   b. 5 units/hour.
   c. 4 units/hour.
   d. 3 units/hour.

   Answer: _____

4. [2 points] A retailer faces daily demand with std dev = 25 units. Lead time = 4 days (constant). For a 95% service level (z = 1.96), what is the required safety stock?
   a. 49 units.
   b. 98 units.
   c. 196 units.
   d. 200 units.

   Answer: _____

5. [2 points] A food vendor buys hot dogs at $2 and sells them at $5. Unsold hot dogs are thrown away (zero salvage). What is the critical fractile?
   a. 0.40.
   b. 0.50.
   c. 0.60.
   d. 0.71.

   Answer: _____

6. [2 points] A supermarket has two checkout lanes, each with its own queue, both at 70% utilization. The manager pools them into one queue with two servers. What is the utilization of the pooled system?
   a. 35%.
   b. 70%.

c. Above 70%, because all customers now share one queue.
d. Cannot be determined without knowing arrival rates explicitly.

Answer: _____

7. [2 points] A firm's fixed ordering cost K is reduced to K/4. All other EOQ parameters remain the same. The new optimal Q* is:
   a. One-quarter of the original Q*.
   b. One-half of the original Q*.
   c. Twice the original Q*.
   d. Four times the original Q*.

   Answer: _____

8. [2 points] In the Toyota Production System, poka-yoke refers to making an activity 'foolproof.' What is its primary operational effect?
   a. It reduces the variability of processing times at that step.
   b. It reduces the amplitude of demand fluctuations passed to upstream processes.
   c. It empowers workers to stop the production line whenever a quality defect is detected.
   d. It makes the amount of work-in-process inventory more readily visible.

   Answer: _____

Question 2 — ClinicFlow Urgent Care (Queueing) [9 POINTS]

ClinicFlow Urgent Care has 3 exam rooms (each staffed by one physician). Patients arrive at 30/hr (CV_a = 1.0). Average consultation = 4 min (CV_s = 1.0).

a) [3 pts] What is the utilization of each exam room? Show your work.

b) [3 pts] ClinicFlow is considering opening a 4th exam room. What will happen to (i) utilization per room, and (ii) average patient wait time? Justify briefly with reference to the queueing formula.

c) [3 pts] A consultant proposes standardizing consultations to reduce CV_s from 1.0 to 0.5 (mean service time unchanged). Without calculation, explain the direction and mechanism of change in average wait time.

Question 3 — LuminaPack Packaging (Process Analysis) [18 POINTS]

LuminaPack — five-step serial process (A–E). Assume no variability; each worker is dedicated to one step.

Step	A	B	C	D	E
Task time (min/unit)	10	50	20	30	15
Number of workers	1	2	1	2	2

For questions 3.1–3.5, circle the correct answer. For 3.6, show your work.

3.1. [2 pts] What is the minimum possible throughput time for the LuminaPack process?

a. 10 min.
b. 50 min.
c. 95 min.
d. 125 min.
e. None of the above.

Answer: _____

3.2. [2 pts] What is the steady-state process capacity for LuminaPack?

a. 1.2 u/hr
b. 1.5 u/hr
c. 2.0 u/hr
d. 2.4 u/hr
e. 3.0 u/hr
f. None of the above

Answer: _____

3.3. [2 pts] Which step is the bottleneck in LuminaPack?

a. Step A.
b. Step B.
c. Step C.
d. Step D.
e. Step E.

Answer: _____

3.4. [3 pts] One worker moves from Step E to Step B (B now has 3 workers, E has 1). What is the new process capacity?

a. 2.4 u/hr
b. 3.0 u/hr
c. 3.6 u/hr
d. 4.0 u/hr
e. None of the above

Answer: _____

3.5. [3 pts] With all 8 workers fully flexible (original config), what is the minimum number of workers needed to achieve at least 3 u/hr capacity?

a. 3 workers.
b. 4 workers.
c. 5 workers.
d. 6 workers.
e. 7 workers.

Answer: _____

3.6. [6 pts] Process redesign into 3 steps: W = 25, X = 50, Y = 25 min/unit. Allocate all 8 original workers to maximize capacity. (a) Describe your allocation strategy and (b) calculate the resulting capacity (u/hr).

Question 4 — WaveCrest Surfboards (Inventory) [21 POINTS]

WaveCrest Surfboards — limited-edition summer model (single order, no reorder).

Parameter	Value
Manufacturing cost	$100/board
Full-price sale	$400/board
Closeout sale	$25/board
Demand distribution	Normal, mean = 150, std dev = 50 boards

Standard Normal CDF — use smallest z where P(Z ≤ z) ≥ CF:

z	P(Z ≤ z)	z	P(Z ≤ z)
0.0	0.500	1.1	0.864
0.1	0.540	1.2	0.885
0.2	0.579	1.3	0.903
0.3	0.618	1.4	0.919
0.4	0.655	1.5	0.933
0.5	0.691	1.6	0.945
0.6	0.726	1.7	0.955
0.7	0.758	1.8	0.964
0.8	0.788	1.9	0.971
0.9	0.816	2.0	0.977
1.0	0.841		

a) [3 pts] What are WaveCrest's C_o and C_u? Explain each.

b) [4 pts] Find the critical fractile and optimal Q*. Show your work using the z-table above.

c) [3 pts] With Q* = 195, what is the probability that WaveCrest will have unsold boards at season's end?

d) [4 pts] A surf shop offers to buy any unsold boards at $55 each instead of $25. How does this change Q*? Use the z-table above and explain why.

e) [3 pts] WaveCrest considers switching to a domestic manufacturer allowing a mid-season reorder. How would this change the initial order quantity and expected profit? No calculation needed.

f) [4 pts] WaveCrest also buys surfboard wax: annual demand 1,800 units, fixed order cost $50, annual holding cost $2/unit. (i) EOQ? (ii) Orders per year?

Question 5 — Miscellaneous Open-Text Questions [21 POINTS]

Each sub-question is worth 3 points.

a) [3 pts] If managing your email inbox is your personal bottleneck, describe three strategies to manage it better and link each to an operations management concept.

b) [3 pts] A manager argues: "Our agents are idle 20% of the time, so we can cut the workforce by 10% without affecting wait times." Is this correct? Explain using queueing logic.

c) [3 pts] Hospital Alpha has highly predictable patient volume (low CV). Hospital Beta serves a walk-in population with high variability (high CV). Both operate at 80% utilization. Which hospital will have longer average wait times, and why?

d) [3 pts] A clothing retailer orders its flagship jeans twice a year in large shipments. A supply chain consultant suggests ordering quarterly instead. What are the tradeoffs? Under what conditions is quarterly ordering better?

e) [3 pts] A concert venue must decide how many VIP parking passes to print before each show. Passes sell for $50 and cost $5 to print; unsold passes are worthless. Using the newsvendor framework, explain what the optimal printing quantity depends on.

f) [3 pts] Toyota's "andon cord" allows any worker to signal a problem that can stop part or all of the production line. What operational principle does this embody, and what are two practical consequences for quality and efficiency?

g) [3 pts] The "Good Jobs Strategy" argues paying workers well and investing in their training improves operational performance. Explain the mechanism — why does paying workers more lead to better operations?

Queueing theory formulas

Here are the queueing formulas, first the general approximations, and in the box to the right the special cases with m=1 server and with coefficients of variation of interarrival times and service times $CV_{IAT} = CV_{ST} = 1$.
Notation:

- A is the arrival rate (in customers per unit time)
- S is the service rate (in customers per unit time)
- m is the number of servers
- u is the utilization, defined as u = A/(m x S)
- CV_{IAT} : coefficient of variation of interarrival times = standard deviation / mean of interarrival times
- CV_{ST} : coefficient of variation of service times = standard deviation / mean of service times

- Average number in line $$L_q \approx \frac{u^{\sqrt{2(m+1)}}}{(1-u)} \times \frac{(CV_{IAT})^2 + (CV_{ST})^2}{2}$$
- Average time waiting in line $$W_q \approx \frac{1}{A} \times \frac{u^{\sqrt{2(m+1)}}}{(1-u)} \times \frac{(CV_{IAT})^2 + (CV_{ST})^2}{2}$$
- Average time in system $$W \approx \frac{1}{A} \times \frac{u^{\sqrt{2(m+1)}}}{(1-u)} \times \frac{(CV_{IAT})^2 + (CV_{ST})^2}{2} + \frac{1}{S}$$
- Average number in system $$L \approx \frac{u^{\sqrt{2(m+1)}}}{(1-u)} \times \frac{(CV_{IAT})^2 + (CV_{ST})^2}{2} + \frac{A}{S}$$

$$L_q = \frac{u^2}{1-u}$$

$$W_q = \frac{1}{S} \times \frac{u}{1-u}$$

$$W = \frac{1}{S} \times \frac{1}{1-u}$$

$$L = \frac{u}{1-u}$$

Inventory theory formulas

Here are the formulas we introduced for the various inventory models.
Economic Order Quantity (EOQ) model
Definitions:

- D = yearly demand rate (units/year)
- h = unit inventory holding costs ($/unit/year)
- K = fixed ordering cost (or set up cost) ($/order)
- Q = order quantity (units)

Optimal order quantity is the Economic Order Quantity:

- $Q^* = \sqrt{\frac{2\,K\,D}{h}}$

Newsvendor model
Definitions:

- D = (Random) Demand of mean μ and std dev σ.
- Probability distribution (e.g., Normal): $prob(D \leq Q)$
- Co = Overage Cost
- Cu = Underage Cost
- Q = Order Quantity
- critical fractile = cu/(co+cu)
- for given order quantity Q, the service level = $prob(D \leq Q)$

The newsvendor model prescribes that you should choose an optimal order quantity Q* such that the service level = the critical fractile, i.e., $prob(D \leq Q)$ = cu/(co+cu).
What is the service level (SL) if we know:

- the quantity Q? SL = NORM.DIST(Q, mean, stdev, TRUE)
- the z-score? SL = NORM.S.DIST(z, TRUE)

Given a service level (SL):

- what is the quantity Q? Q(SL) = NORM.INV(SL, mean, stdev)
- what is the z-score? z(SL) = NORM.S.INV(SL)

Need to find optimal Q^$*$ such that the service level satisfies:
$prob(D \leq Q$^$*)$ = NORM.DIST (Q^$*$, m, s, TRUE) = $cu/(cu+co$)
Use the inverse of the normal distribution:
$Q*$ = NORM.INV($cu/(cu+co$), m, s, TRUE)
Or use the inverse of the standard normal distribution:
z$*$= NORM.S.INV($cu/(cu+co$),TRUE)
Q^$*$ = μ + $\sigma \times z*$

Reorder point model
Definitions:

- D = (Random) Demand of mean μ and std dev σ. (units/period)
- Probability distribution (e.g., Normal): $prob(D \leq Q)$
- h = unit inventory holding costs ($/unit/period)
- K = fixed ordering cost (or set up cost) ($/order)
- LT = lead time (between placing an order and receiving the order) (periods)
- Q = Order Quantity
- target service level SL; for given order quantity Q, the service level = $prob(D \leq Q)$
- the z-score corresponding to service level SL = z_{SL}

The reorder point model has two parameters: the order quantity Q, and the reorder point ROP.
For the order quantity Q, use the EOQ formula.
For the reorder point: ROP = $\mu \times LT + z_{SL} \times \sigma_D \times \sqrt{LT}$

Process analysis definitions

- Cycle Time: Time between completion of successive units (e.g., minutes / patient, hours / customer)
- Capacity: How much can be processed per unit time (e.g., patients / minute, customers / hour)
- Flow Rate: How much is actually processed per unit time
    - Flow rate = minimum{Demand, Process Capacity}
- Utilization: proportion (or %) of capacity used
    - Utilization = Flow Rate / Capacity , OR
    - Utilization = Time Occupied / Time Available
- Bottleneck: Resource which limits the flow
    - Often (in simple processes): bottleneck is step with the lowest capacity, or step with highest utilization
- Balance/Imbalance: Balanced process if all stages have equal capacity
- Idle Time (IT): Time spent per cycle not doing useful work
- Direct Labor Content (DLC): Total operator time spent per unit of product (i.e., per cycle)
- Direct Labor Utilization (DLU): Percentage of operator time spent doing useful work
    - DLU = DLC / (DLC + IT)
    - DLU = DLC / (total operator time per cycle)
- Throughput Time (TT): Sum of processing and wait times at all stages of the process
- Lot (Batch) Size: Number of units of the same product produced together before starting a new batch
- Setup Time: Preparation time before batch is produced
- Run Time: Actual time to produce the batch, excluding the setup time